INCH-POUND

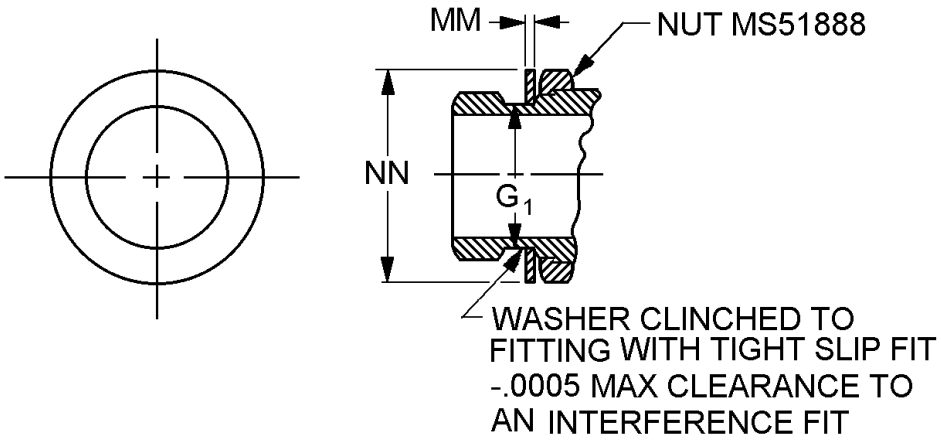
MS51889A
w/AMENDMENT 1
24 September 2018
SUPERSEDING
MS51889A
24 May 2018

DETAIL SPECIFICATION SHEET

WASHER, BACK-UP, 37 DEGREE FLARED AND FLARELESS TYPE, HYDRAULIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.



Inches	mm
.0005	0.013

See notes at end of figure.

FIGURE 1. Washer, backup.

AMSC N/A

FSC 4730

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Tube OD Nom inches (mm)	G_1 Diameter .002 (0.05) -.003 (0.13)	MM inches (mm) ±.003 (0.13)	NN Diameter ±.015 (0.38)
.1250 (3.175)	.250 (6.35)	.030 (0.76)	.504 (12.80)
.1875 (4.763)	.313 (7.95)	.030 (0.76)	.575 (14.61)
.2500 (6.350)	.364 (9.25)	.035 (0.89)	.650 (16.51)
.3125 (7.936)	.427 (10.85)	.035 (0.89)	.722 (18.34)
.3750 (9.525)	.482 (12.24)	.035 (0.89)	.794 (20.17)
.5000 (12.700)	.660 (16.76)	.041 (1.04)	1.010 (25.65)
.6250 (15.875)	.773 (19.63)	.050 (1.27)	1.155 (29.34)
.7500 (19.050)	.945 (24.00)	.050 (1.27)	1.444 (36.68)
.8750 (22.225)	1.070 (27.18)	.050 (1.27)	1.589 (40.36)
1.0000 (25.400)	1.195 (30.35)	.050 (1.27)	1.732 (43.99)
1.2500 (31.750)	1.507 (38.28)	.050 (1.27)	2.165 (54.99)
1.5000 (38.100)	1.756 (44.60)	.050 (1.27)	2.454 (62.33)
2.000 (50.800)	2.381 (60.48)	.050 (1.27)	3.160 (80.26)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Washer shall be free from all hanging burrs, loose scale, or slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514.
5. Threads shall be in accordance with ASME B1.1.
6. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.

FIGURE 1. Washer, backup - Continued.

REQUIREMENTS:

Washer shall be as specified on figure 1 and in table I.

This is a design standard and shall not be used for procurement.

Used on MS51527, MS51528, MS51529, MS51530, MS51839, MS51841, and MS51842.

Materials shall be in accordance with MIL-DTL-18866.

Finish. Finishes shall be as specified in table I. All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The washers shall show no evidence of red corrosion after 96 hours of salt spray.

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TABLE I. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/
H	Steel	Zinc-aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 2/
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/ 3/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 4/
R	Steel	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/ 5/
SS	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7, free machining alloy UNS S30300 method 1, type 2 or 4.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 6/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 6/
Z	Steel	Zinc plating in accordance with ASTM B695, type II, class 8. 4/ 5/
ZC	Steel	Zinc may be any zinc plating from PIN codes J, R, and Z with a colored chromate coating. 4/ 5/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA owned shipboard systems.
2/ Hexavalent chromium free. Finish shall be RoHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications.
5/ These finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended.
6/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293. Titanium shall not be used in oxygen or potable water systems.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2.

Class I and II ozone depleting substances (ODS) shall not be used in MS51889 or any referenced procedures.

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Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

This is a design standard and shall not be used as a Part or Identifying Number (PIN).

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

MIL-DTL-16232	ASTM F1136/F1136M	SAE-AMS-STD-595/36251
MS51527	SAE-AMS-C-81562	SAE-AMS-STD-595/36270
MS51528	SAE-AMS-QQ-P-416	SAE-AMS-STD-595/36280
MS51529	SAE-AMS-STD-595/36076	SAE-AMS-STD-595/36293
MS51530	SAE-AMS-STD-595/36081	SAE-AMS2417
MS51839	SAE-AMS-STD-595/36099	SAE-AMS2486
MS51841	SAE-AMS-STD-595/36118	SAE-AMS2488
MS51842	SAE-AMS-STD-595/36134	SAE-AMS2700
MS51888	SAE-AMS-STD-595/36152	SAE-J514
ASME B1.1	SAE-AMS-STD-595/36170	T9070-AL-DPC-020/077-2
ASTM B117	SAE-AMS-STD-595/36173	
ASTM B633	SAE-AMS-STD-595/36176	
ASTM B695	SAE-AMS-STD-595/36231	

CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - OS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2018-088)

Review activities:
Army - AT, GL, MI
Navy - AS
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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